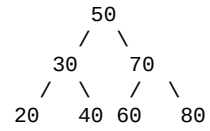
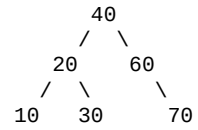
RES POLYTECHNIC, BANGALORE-91

CIE Theory Test III

Program	Computer Science and Engineering		Semester- III
Course Name	Data Structure with Python		Test III
Course Code	25CS33I	Duration- 90 min	Marks-50
Name of the Course Coordinator: Vinay Kumar			
Note: Answer any one full question from each section. Each full question carries equal marks.			

Q. No	Questions	CL	CO	Marks
	SECTION I			
1	**a)** Define a tree. For the binary tree given below, identify the root node, the leaf nodes, the parent of node 40, the children of node 30 and the height of the tree. Write any two properties of trees. **(5 marks)**  **b)** Explain how recursion works using stack memory and execution flow. Write a recursive Python function to find the factorial of a number and trace it for n = 4, showing the base case and the recursive case. Differentiate between recursion and iteration. **(10 marks)** **c)** Write a Python program that reads two numbers from the user and displays their division. Use exception handling to show suitable messages for division by zero and for invalid (non-numeric) input, and display "Program completed" using the finally block. **(10 marks)** **Example** Input: 10 and 0 Output: Cannot divide by zero Program completed	L2, L3	1, 3, 5	25
	OR			
2	**a)** Write the In-order, Pre-order and Post-order traversals of the binary tree given below. **(5 marks)**  **b)** What are the types of errors in Python? Write a Python program that takes a list of numbers and an index from the user and displays the element at that index. Use exception handling to handle IndexError and ValueError. **(10 marks)** **Example** Input: [10, 20, 30], index = 5 Output: Index is out of range **c)** Define an Abstract Data Type (ADT) and a User-Defined Data Type (UDT). List any four common ADTs. Write a Python class Stack (UDT) with push() and pop() methods to show how a UDT is used to implement the Stack ADT. **(10 marks)**	L2, L3	1, 2, 3, 5	25

Q. No	Questions	CL	CO	Marks
	SECTION II			

3	Analyze the code and write the output.
	a) (10 marks)
	`class Stack:`
	`  def __init__(self):`
	`  self.items = []`
	`  def push(self, x):`
	`  self.items.append(x)`
	`  def pop(self):`
	`  if self.items == []:`
	`  return "Stack Underflow"`
	`  return self.items.pop()`
	`  def peek(self):`
	`  return self.items[-1]`
	`  def is_empty(self):`
	`  return len(self.items) == 0`
	
	`s = Stack()`
	`s.push(10)`
	`s.push(20)`
	`s.push(30)`
	`print(s.pop())`
	`print(s.peek())`
	`s.push(40)`
	`print(s.pop())`
	`print(s.pop())`
	`print(s.pop())`
	`print(s.is_empty())`
	`print(s.pop())`
	b) (15 marks)
	`class Node:`
	`  def __init__(self, data):`
	`  self.data = data`
	`  self.next = None`
	
	`class LinkedList:`
	`  def __init__(self):`
	`  self.head = None`
	`  def insert_begin(self, data):`
	`  new = Node(data)`
	`  new.next = self.head`
	`  self.head = new`
	`  def insert_end(self, data):`
	`  new = Node(data)`
	`  if self.head is None:`
	`  self.head = new`
	`  return`
	`  temp = self.head`
	`  while temp.next:`
	`  temp = temp.next`
	`  temp.next = new`
	`  def delete_value(self, key):`
	`  temp = self.head`
	`  if temp and temp.data == key:`
	`  self.head = temp.next`
	`  return`
	`  prev = None`
	`  while temp and temp.data != key:`
	`  prev = temp`
	`  temp = temp.next`
	`  if temp is None:`
	`  print(key, "not found")`
	`  return`
	`  prev.next = temp.next`
	`  def display(self):`
	`  temp = self.head`
	`  while temp:`
	`  print(temp.data, end=" -> ")`
	`  temp = temp.next`
	`  print("None")`
	
	`ll = LinkedList()`
	`ll.insert_end(10)`
	`ll.insert_end(20)`
	`ll.insert_begin(5)`
	`ll.display()`
	`ll.insert_end(30)`
	`ll.delete_value(20)`
	`ll.display()`
	`ll.delete_value(99)`
	`ll.delete_value(5)`
	`ll.display()`
		L3	1, 3, 4	25

Q. No	Questions	CL	CO	Marks
	OR			
4	**a)** Create a Python program to implement a Queue using a class with the operations enqueue(), dequeue() (display a message when the queue is empty), display() and is_empty(). **(12 marks)** Enqueue the customer names Asha, Ravi, Meena and Kiran, dequeue two customers and display the remaining queue with relevant messages. Suggest the suitable Python concept to solve the above and justify your selection. **b)** Create a Python program to implement a Doubly Linked List using a Node class having previous and next pointers. **(13 marks)** 1. Insert nodes 10, 20, 30 at the end and node 5 at the beginning. 2. Delete the node having the value 20. 3. Traverse and display the list in the forward direction and in the backward direction. Suggest the suitable Python concept to solve the above and justify your selection.	L3	1, 3, 4	25

Note for the Course Coordinator: Each question may have one, two or three subdivisions. Optional questions in each section carry the same weightage of marks, cognitive level and course outcomes.

Sign of the Course Coordinator Signature of the HOD Signature of the IQAC Chairman